

KIRTAN KUMAR PATEL

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Education

VIT Bhopal University

B.Tech in Computer Science specialization in AI/ML – CGPA: 8.4

09/2023 – 05/2027

Bhopal, MP

St. Gabriel's Hr. Sec. School

Class XII (CBSE) – 84%

2022

Jabalpur, MP

St. Gabriel's Hr. Sec. School

Class X (CBSE) – 94%

2020

Jabalpur, MP

Technical Skills

Languages: Python, SQL

AI & Machine Learning: RAG, Sentence Transformers, Cross-Encoders, Scikit-Learn, YOLO, OpenCV

Cloud & MLOps: AWS (EC2, S3, ECR, IAM), Docker, GitHub Actions (CI/CD)

Backend & Databases: FastAPI, PostgreSQL, MongoDB Atlas, Redis, Qdrant (Vector DB)

Projects

Cat RAG – Secure Multi-Tenant RAG Assistant | *Python, FastAPI, Qdrant, PostgreSQL, Redis, Docker*

- Built a session-isolated RAG backend supporting persistent portfolio knowledge and temporary user-uploaded documents, with SHA-256 based single-ingestion, metadata filtering, server-side TTL cleanup, and source-level citations.
- Implemented hybrid retrieval using dense vector search + BM25, Reciprocal Rank Fusion (RRF), conditional query rewriting, and local cross-encoder reranking to improve retrieval precision while minimizing external inference.
- Engineered zero-cost guardrails and provider resilience with application-level daily/rolling token budgets, per-session quotas, concurrency controls, provider routing, circuit breakers, fallback providers, and Redis-based caching.
- Added security and evaluation infrastructure covering upload validation/timeouts, prompt-injection defenses, rate limiting, cross-session isolation, retrieval/generation/citation metrics, operational observability, automated tests, and Docker Compose deployment.

Vehicle Insurance MLOps Pipeline | *Python, Scikit-learn, MongoDB Atlas, AWS S3, FastAPI, Docker, GitHub Actions*

- Built an end-to-end MLOps pipeline covering MongoDB-based data ingestion, schema validation, feature transformation, Random Forest training, model evaluation, and cloud artifact management.
- Implemented reusable preprocessing with ColumnTransformer, StandardScaler, MinMaxScaler, one-hot encoding, and SMOTEENN, with persisted transformation artifacts for inference.
- Developed a model promotion workflow that compares the newly trained model against the existing S3 model using F1-score and pushes the new model when performance improves.
- Containerized the FastAPI prediction application with Docker and automated cloud deployment through GitHub Actions, AWS ECR, and EC2.

Certifications

AIML (Specialization in Drone Technology)

June 2025 – Dec 2025

TiHAN, IIT Hyderabad

- Learning applications of Artificial Intelligence and Machine Learning in autonomous systems and drone technology.